

Name Key

Physics 390 In-Class Activity 9

The 13-Microsecond Rule

The "13-microsecond" rule is a handy way to approximate depths of soft tissues from ultrasound images.

- (A) Verify this rule by showing that every additional 13 μs of go-return time corresponds to an additional 1 cm of reflector depth. Assume that the speed of sound is 1540 m/s.

The 13 Microsecond Rule

- In soft tissue, every 13 μs of go-return time means the reflector is 1 cm deeper in the body.

Time of Flight	Reflector Depth	Total Distance Traveled
13 μs	1 cm	2 cm
26 μs	2 cm	4 cm
130 μs	10 cm	20 cm

$$\text{distance} = \text{speed} \times \text{time}$$

$$2d_1 = v \times t_1$$

$$2d_2 = v \times t_2$$

$$2(d_2 - d_1) = v \times (t_2 - t_1) \Rightarrow 2\Delta d = v \times \Delta t$$

$$\Delta d = 1540 \times \frac{13 \times 10^{-6}}{2} = 10^{-2} \text{ m} = 1 \text{ cm}$$

$$\text{distance} = 2d \leftarrow \text{depth}$$

$$2d = v \times t$$

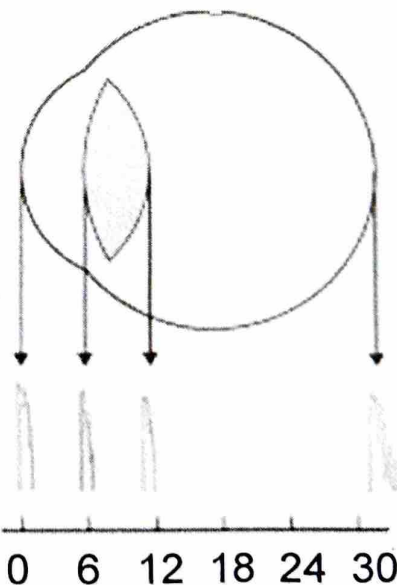
- (B) Use this rule to estimate the thickness of the lens (shown in green) at right.

The two reflections from the surfaces of the lens are 12 μsec - 6 μsec = 6 μsec apart.

$$\text{Since } 13 \mu\text{sec} \leftrightarrow 1 \text{ cm}$$

$$6.5 \mu\text{sec} \leftrightarrow 0.5 \text{ cm}$$

Lens is a little under 0.5 cm thick



μs